

EXPERIEMNT 21

Prolog Program: Towers of Hanoi

Aim

To implement the **Towers of Hanoi problem using Prolog** and display the sequence of moves required to transfer all disks from the source peg to the destination peg.

Algorithm

1. Start with N disks on the source peg.
2. If there is only one disk, move it directly from source to destination.
3. For more than one disk:
 - Move N-1 disks from source to auxiliary peg.
 - Move the largest disk from source to destination.
 - Move N-1 disks from auxiliary peg to destination.
4. Repeat until all disks are moved.
5. Display each move.

Prolog Code

% Towers of Hanoi

hanoi(1, Source, Destination, _):-

write('Move disk from '),

write(Source),

write(' to '),

```
write(Destination),
```

```
nl.
```

hanoi(N, Source, Destination, Auxiliary) :-

```
    N > 1,
```

```
    M is N - 1,
```

```
    hanoi(M, Source, Auxiliary, Destination),
```

```
    write('Move disk from '),
```

```
    write(Source),
```

```
    write(' to '),
```

```
    write(Destination),
```

```
    nl,
```

```
    hanoi(M, Auxiliary, Destination, Source).
```

Input

```
hanoi(3, left, right, middle).
```

Output

```
(16 ms) yes
| ?- hanoi(3, left, right, middle).
Move disk from left to right
Move disk from left to middle
Move disk from right to middle
Move disk from left to right
Move disk from middle to left
Move disk from middle to right
Move disk from left to right
true ? |
```

Result

Thus, the Towers of Hanoi problem was successfully implemented using Prolog, and the sequence of moves required to transfer the disks from the source peg to the destination peg was obtained.